

Literature Status: Two Answered Problems from *J-class operators and hypercyclicity*

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Status

This is a lightweight literature-status packet for two specific questions from Costakis–Manoussos, *J-class operators and hypercyclicity*, arXiv:0704.3354 [1]. It does not settle all five open problems in the source paper.

Source Questions

In Section 6, page 19 of the source PDF, the authors ask:

Problem 2. Does every non-separable and infinite dimensional Banach space support a J -class operator?

They then ask:

Problem 3. Is there a spectral description of the closure of the set of J -class operators acting on a Hilbert space?

Supporting Answer to Problem 2

Nasseri, *On the existence of J -class operators on Banach spaces*, arXiv:1009.3461 [2], explicitly answers the Costakis–Manoussos non-separable-space question in the negative. The abstract states that the paper answers the question negatively, and Theorem 1.1 states that there exists a non-separable complex Banach space X on which the J -set of every operator has empty interior for every nonzero vector. Consequently X supports no J -class operator.

This is a direct negative answer to Problem 2.

Supporting Answer to Problem 3

Tian–Hou, *Limits of J -class operators*, arXiv:1010.3386 [3], explicitly addresses the third Costakis–Manoussos problem. Its abstract says that it gives the spectral description of the closure of the set of J -class operators on a separable Hilbert space. In Section 1 it repeats the question, and Theorem 2.10 gives the announced formula:

$$\begin{aligned}\overline{J(H)} &= \{T \in B(H) : [\sigma_{\text{re}}(T) \cup \rho_{\text{s-F}}^{(+)}(T)] \cap \partial\mathbb{D} \neq \emptyset, \\ &\quad \overline{\mathbb{D}} \cap \rho_{\text{s-F}}^{(-)}(T) = \emptyset, \\ &\quad \text{no component of } \sigma(T) \text{ is contained in } \mathbb{D}\}.\end{aligned}$$

This is a direct literature answer to Problem 3.

Scope Limitations

Problems 1, 4, and 5 from the same source open-problems section are not answered by this packet. A bounded direct attack checked the natural direct-sum route for Problem 4, but individual hypercyclicity of summands is not enough because J -sets of direct sums require common return times. No promoted proof or counterexample was obtained for Problems 1, 4, or 5.

Search Notes

The run indexes had no exact packet for arXiv:0704.3354. They did contain a nearby triage entry for arXiv:1009.3461, recording that Nasserri answers the non-separable J -class existence question in the negative. Local source search also found arXiv:1010.3386, whose abstract and introduction explicitly identify Problem 3 from Costakis–Manoussos and whose Theorem 2.10 supplies the spectral description.

References

- [1] G. Costakis and A. Manoussos, *J-class operators and hypercyclicity*, arXiv:0704.3354.
- [2] A. B. Nasserri, *On the existence of J-class operators on Banach spaces*, arXiv:1009.3461.
- [3] G. Tian and B. Hou, *Limits of J-class operators*, arXiv:1010.3386.